

15) Prove that if x is an irrational number and $x > 0$, then $\sqrt{x}$ is also irrational.

Solution: $p: x$ is an irrational number $q: x > 0$ $r: \sqrt{x}$ is an irrational number. We shall prove $(p \wedge q) \rightarrow r$ by contradiction.

$\therefore$ We shall prove $(p \wedge q) \wedge \neg r \rightarrow \text{False}$ is True.

Let $(p \wedge q) \wedge \neg r$ be True. By defn. of AND, $(p \wedge q)$ is True and $\neg r$ is True.

$\neg r: \sqrt{x}$ is not an irrational no., or by defn. 2, $\sqrt{x}$ is a rational number.

By defn. of AND, p is True and q is True. $\therefore x$ is an irrational number and $x > 0$.

By defn. 2, there exists integers a and b , where $b \neq 0$ such that $\sqrt{x} = \frac{a}{b}$. $\therefore (\sqrt{x})^2 = \frac{a^2}{b^2} \Rightarrow x = \frac{a^2}{b^2}$. It's easy to see that

a^2 and b^2 are integers. Since $b \neq 0$, $b^2 \neq 0$.

By defn. 2, x is a rational number. This is a contradiction to the fact that x is an irrational number. $\therefore \neg r$ is False

By defn. of AND, $(p \wedge q) \wedge \neg r$ is False. By defn. of conditional, $(p \wedge q) \rightarrow r$ is True. $\therefore (p \wedge q) \rightarrow r$ is True

$\therefore$ Hence, if x is an irrational no. and $x > 0$, then $\sqrt{x}$ is also irrational $\square$

16) Prove that if x, y and z are integers and $(x+y+z)$ is odd, then at least one of x, y and z is odd.

Solution: $p: x$ is an odd integer $q: y$ is an odd integer $r: z$ is an odd integer $s: (x+y+z)$ is an odd integer

We shall prove that $s \rightarrow (p \vee q \vee r)$ is True by contraposition.

$\therefore$ We shall prove $\neg(p \vee q \vee r) \rightarrow \neg s$ is True.